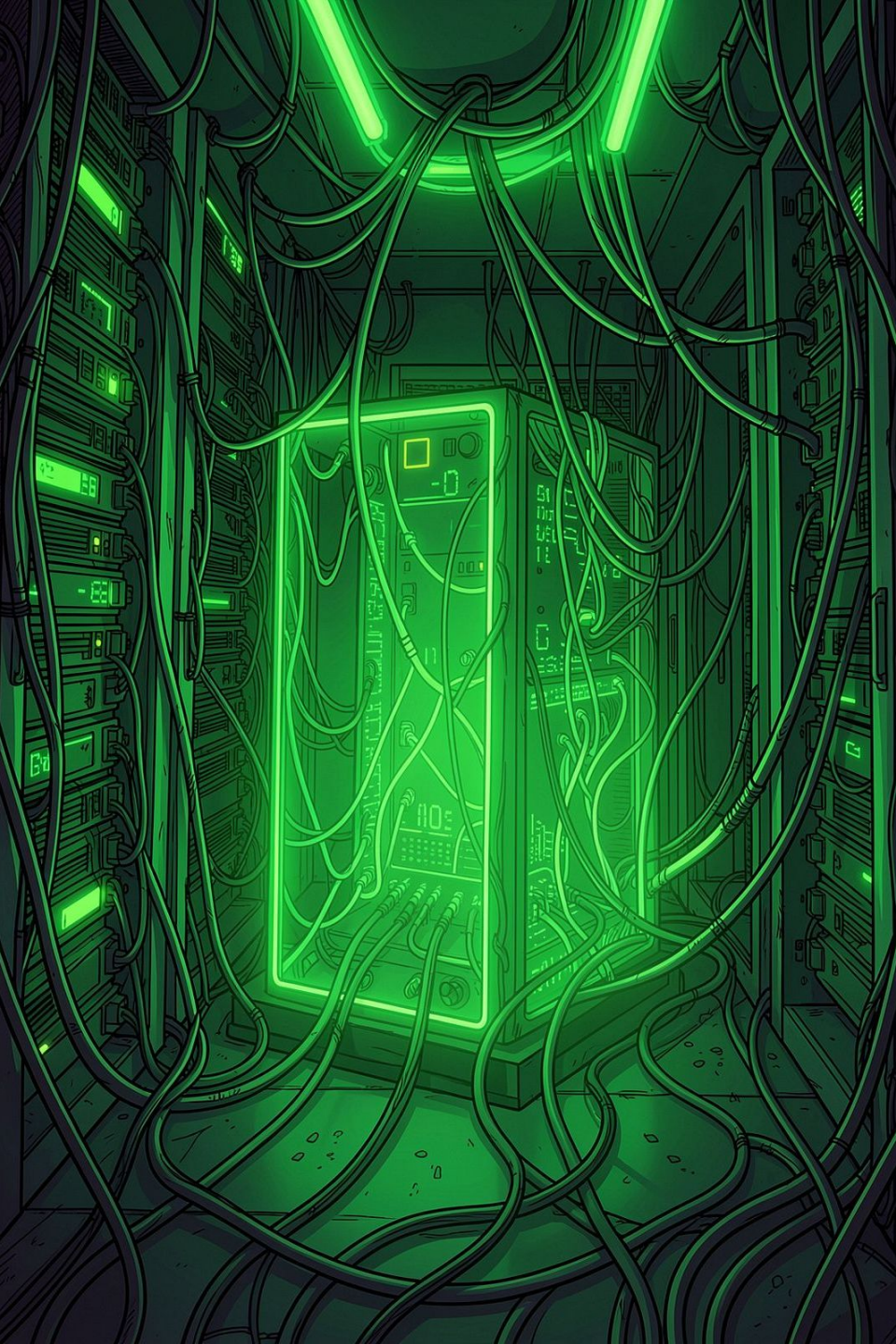
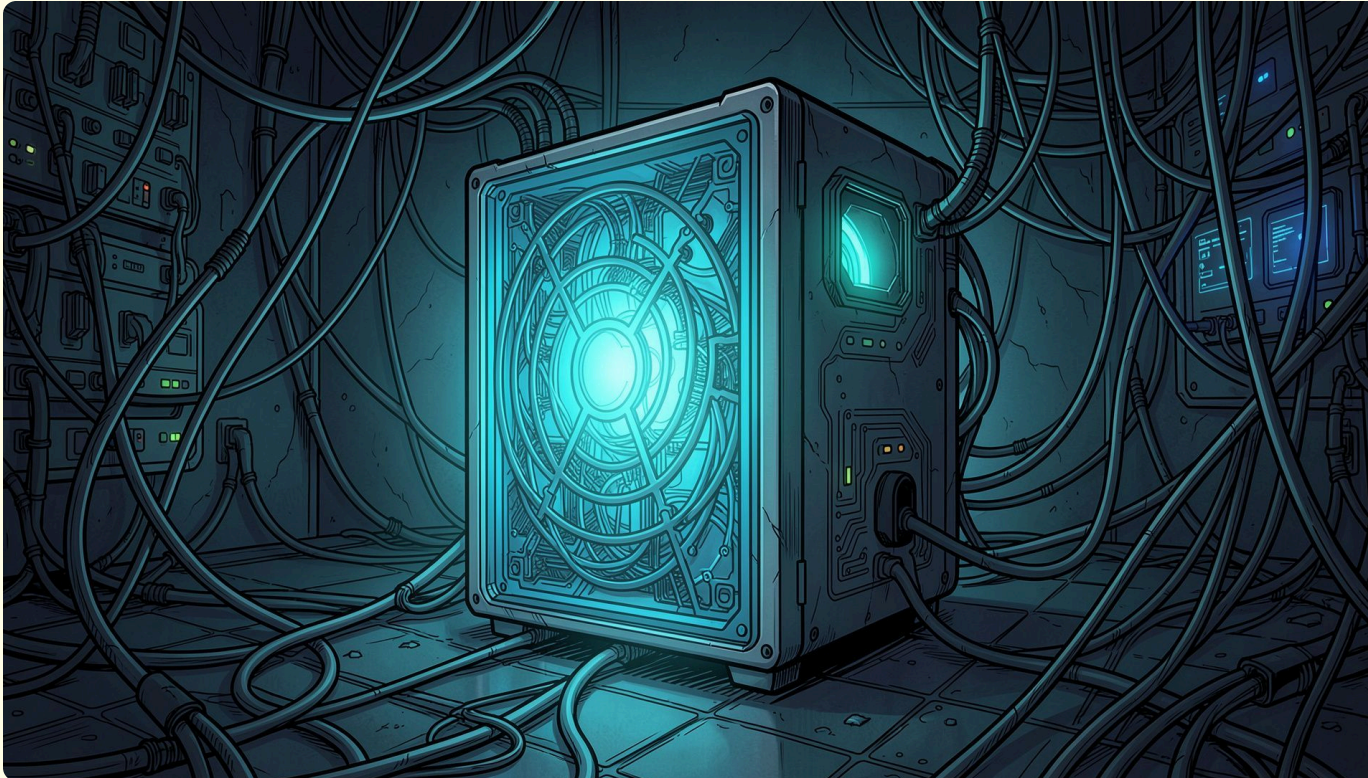




Honeypots in Cybersecurity

A strategic deception tool used to detect, deflect, and study cyber threats — before they reach real assets.

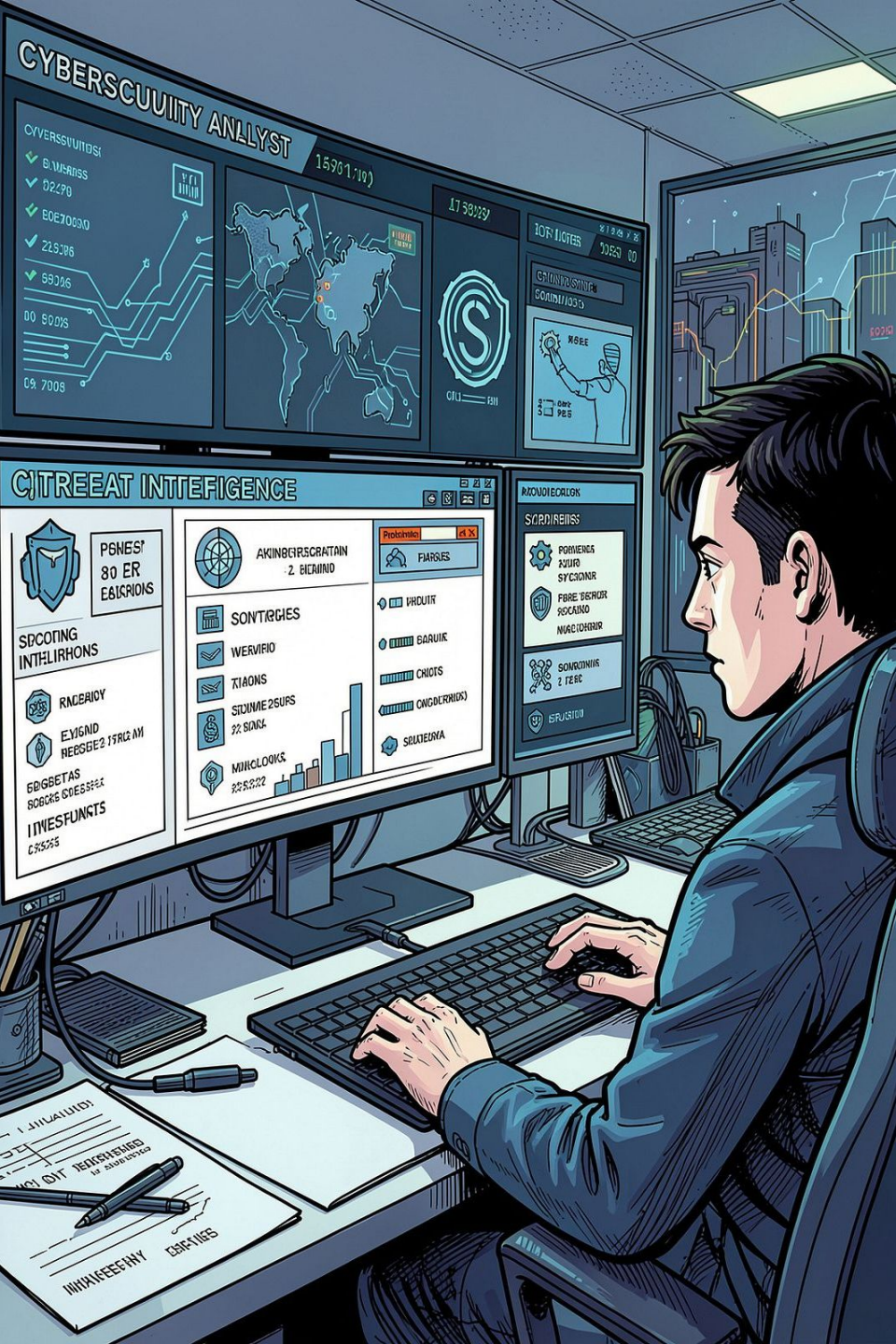
What Is a Honeypot?



A **honeypot** is a security mechanism – typically a fake system, server, application, or network – designed to **mimic a real target** and attract attackers.

Unlike firewalls or IDS that defend, honeypots **invite interaction** to gather intelligence on attacker behavior, tools, and techniques.

- Key principle: Any traffic to a honeypot is, by definition, suspicious – there is no legitimate reason for a user to access it.



Why Honeypots Are Used

Threat Intelligence

Capture real-world attack data – malware samples, TTPs, and attacker infrastructure – to strengthen defenses.

Early Warning

Detect intrusions before they escalate. Honeypots alert teams to active reconnaissance or exploitation attempts.

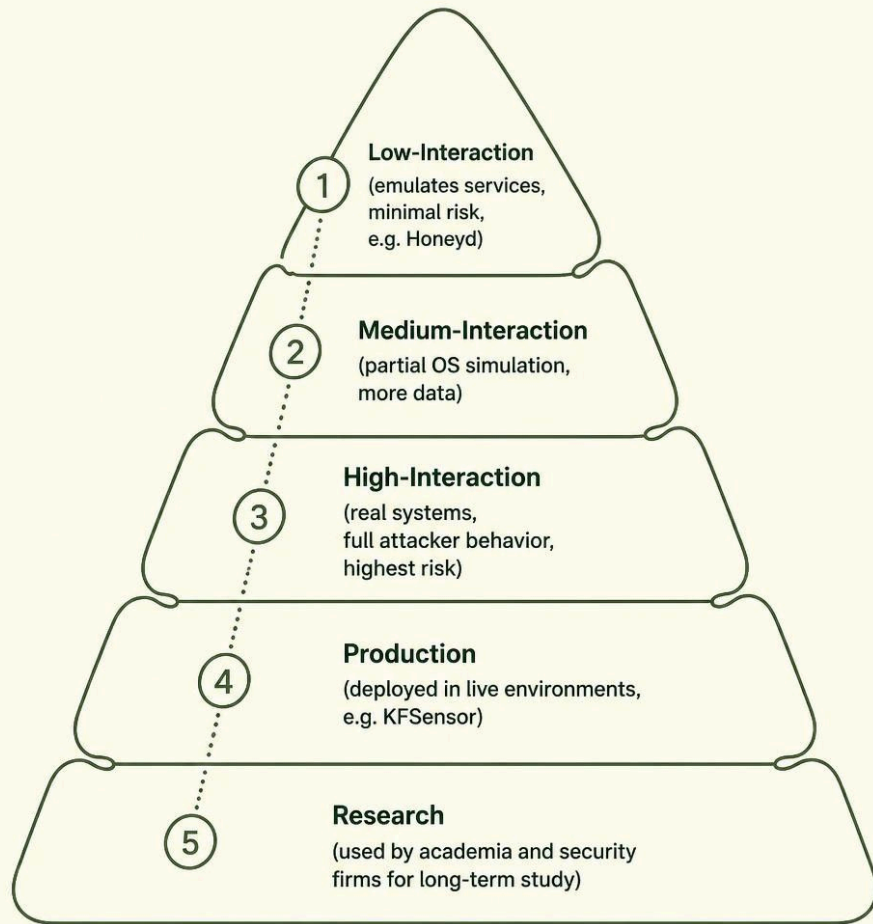
Distraction & Delay

Divert attacker attention from production systems, buying time for defenders to respond.

Research & Training

Study attacker behavior in a controlled environment and train security teams on real threat scenarios.

Types of Honeypots



Honeypots are classified by **level of interaction** and **deployment purpose**:

Low-Interaction

Emulates services – safe, easy to deploy.

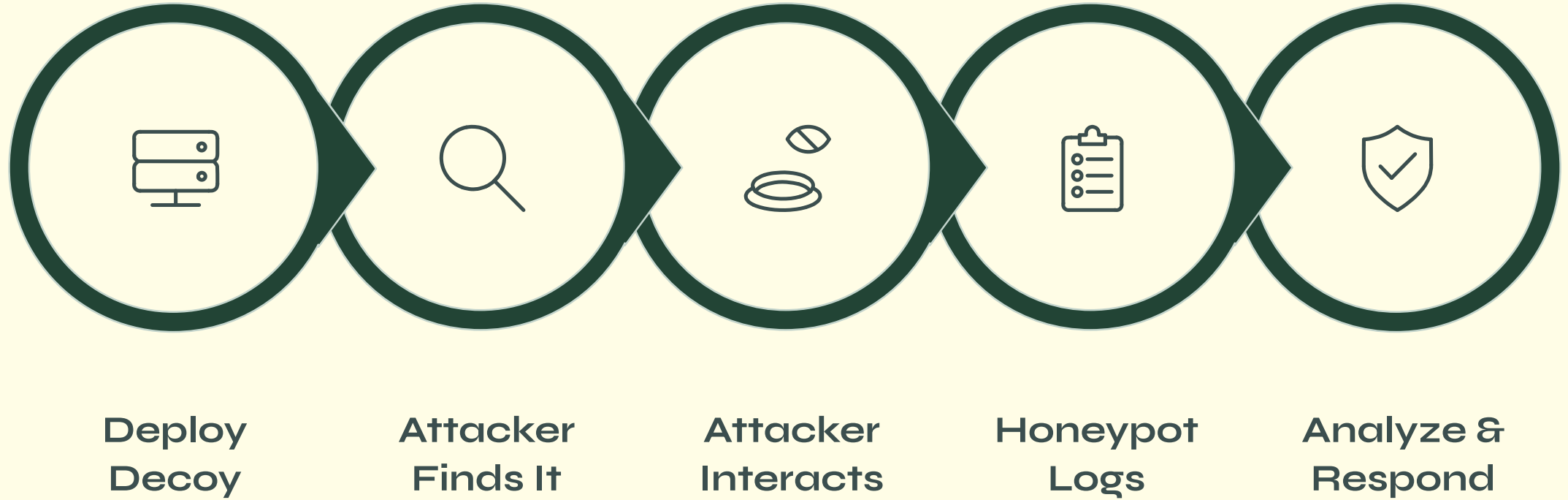
High-Interaction

Real systems that allow full attacker engagement – rich data, higher risk.

Production vs. Research

Production protects live environments; research supports long-term threat study.

How Honey pots Work



The entire interaction is monitored and logged – giving defenders a complete picture of attacker methodology without risking real assets.

Honeypots vs. Honeynets

Honeypot

A **single decoy system** – one server, one service, or one application designed to attract attackers and capture their behavior in a controlled environment.

- Best for monitoring attacks against one exposed asset, such as a fake SSH server, web app, or database
- Lower complexity and easier to deploy, tune, and maintain
- Useful for collecting focused threat intelligence, detecting scanning activity, and studying exploit attempts against a specific target
- Limited scope of data, but often enough for testing alerts, validating defenses, or distracting opportunistic attackers

Honeynet

A **network of multiple honeypots** simulating an entire environment – servers, firewalls, and internal systems – to observe how attackers move, pivot, and escalate once inside.

- Best for mimicking a realistic enterprise environment, such as a small office network or segmented production lab
- Richer attacker behavior data, including lateral movement, privilege escalation, and command-and-control activity
- Can include web servers, file shares, database systems, routers, and fake internal services to make the environment believable
- Requires a honeynet gateway or similar control layer to safely monitor traffic and prevent the environment from being used as a launch point

In practice, a honeypot is ideal when you want depth on one target, while a honeynet is better when you want to understand attacker behavior across an entire simulated environment.

Advantages of Honeypots



Zero False Positives

Any interaction with a honeypot is inherently malicious – eliminating noise common in traditional monitoring.



Deep Threat Visibility

Reveals attacker tools, techniques, and procedures (TTPs) that standard defenses never see.



Reduces Attack Surface

Diverts attackers away from production systems, reducing the likelihood of a successful breach.



Forensic Evidence

Captured data can support incident response, legal proceedings, and threat attribution.



Limitations & Risks

Limited Scope

Honeypots only see traffic aimed at them. They can miss broader attacks elsewhere.

Risk of Compromise

Misconfigured high-interaction honeypots can be exploited. Isolation and monitoring are critical.

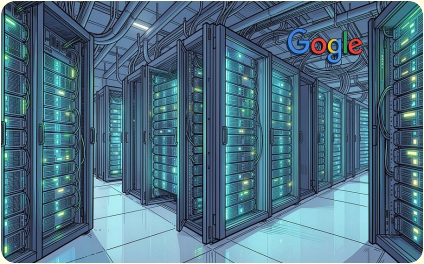
Detection by Attackers

Skilled attackers may spot fake services or odd behavior and avoid them or plant false signals.

Legal & Ethical Concerns

Honeypots can raise privacy, consent, and retention issues. Check regulations before deploying.

Real-World Use Cases



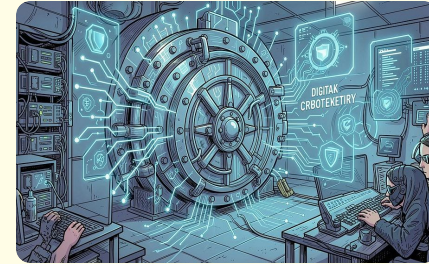
Google's GCP Honeypots

Google deploys honeypots across its cloud infrastructure to detect exploitation attempts and gather threat intelligence at scale.



Microsoft Threat Intelligence

Microsoft uses honeypots to track APT groups, collect malware samples, and feed findings into Defender and Sentinel.



Financial Sector

Banks deploy honeypots to detect lateral movement, credential theft, and insider threats within their networks.

An illustration of three men shaking hands in front of a modern building. The man on the left is wearing a yellow hoodie, the man in the middle is wearing a green jacket, and the man on the right is wearing a light green jacket. They are all smiling. The background shows a modern building with large windows and some trees. The text "Thank You!" is overlaid in the center.

Thank You!